

ESTATEFLOW

Real Estate Marketplace & Lead CRM

Day 1 - Product & Market Analysis

Interview Technical Assignment | Working Draft

1. Executive Summary

EstateFlow is a proposed real-estate marketplace integrated with a lightweight lead-management CRM. The product connects property discovery with the business workflow that follows an enquiry: lead creation, agent ownership, follow-up, site visit, and conversion outcome.

The company reference describes a broader platform with Buyer, Owner, Agent, Builder and Admin users and eight connected areas: Authentication, Property Search, Listings, Maps & Location, Leads & CRM, Site Visits, Projects, and Admin & Reports. For the one-week assignment, the implementation will focus on a coherent MVP that demonstrates the end-to-end business journey rather than attempting every production capability.

2. Problem Statement

A property marketplace can generate enquiries, but a real-estate business must also know who owns each lead, what actions have occurred, when the next follow-up is due, whether a site visit happened, and whether the opportunity converted or was lost. EstateFlow addresses this handoff by making enquiry-to-conversion activity traceable in one system.

3. Users and Core Responsibilities

Role	Primary actions
Buyer	Search, filter, view, shortlist and enquire about properties.
Owner	List and manage properties; respond to interest.
Agent	Work assigned leads, record follow-ups and manage site visits.
Builder	Manage projects/inventory and associated leads at a basic MVP level.
Admin	Verify/monitor users and listings, oversee leads and view reports.

4. Market Research - Initial Competitor View

External research was used to distinguish consumer property marketplaces from specialized real-estate CRM products. The comparison below is intentionally high-level and limited to capabilities visible in official product material.

Product	Primary positioning	Relevant capabilities	Implication for EstateFlow
Magicbricks	Property marketplace / real-estate ecosystem	Property search by location/project, property type and budget; additional filters; owner-posted property filtering.	Strong discovery/search patterns are table stakes.
NoBroker	Brokerage-free property marketplace and related services	Buy/rent/commercial search, owner listings, verified/direct-owner positioning, shortlisting and property services.	Trust, rich listing information and easy owner listing are important.
Sell.Do	Real-estate-specific CRM	Lead capture/management, routing, activity timeline, follow-ups, stage transitions, site-visit management and analytics.	The CRM workflow should be modeled explicitly, not as simple contact-form storage.

5. Proposed Product

Product name (working): EstateFlow

Value proposition: A real-estate marketplace that connects buyer discovery directly to a structured lead pipeline, allowing an enquiry to be followed through agent action, site visit and final conversion outcome.

Core journey: Search -> Property -> Shortlist/Enquiry -> Lead -> Agent -> Follow-up -> Site Visit -> Converted/Lost.

6. MVP Scope - One Week

- Authentication and role-based access for Buyer, Owner, Agent, Builder and Admin.
- Property creation, update, viewing, basic media metadata, publication/verification status.
- Property search and filters such as location, property type, price and bedrooms.
- Buyer shortlist and enquiry creation.
- Automatic or service-layer conversion of an enquiry into a traceable lead.

- Lead assignment, lead stages, notes/activity history and follow-up dates.
- Site-visit scheduling and status management.
- Admin overview for users, properties, leads, visits and basic funnel metrics.
- Validation, error handling, API documentation and basic automated tests.

7. Explicitly Out of MVP / Future Scope

- Payments and booking collections.
- Live RERA or government integrations.
- WhatsApp/SMS/calling infrastructure.
- Advanced AI recommendations or lead scoring.
- Elasticsearch-scale search and advanced geospatial/map search.
- Full builder tower/unit inventory, brokerage accounting and channel-partner management.
- Microservices decomposition, Kubernetes and large-scale production infrastructure.
- Native mobile applications.

8. Proposed Lead Lifecycle

Working lifecycle: NEW -> CONTACTED -> QUALIFIED -> VISIT_SCHEDULED -> FOLLOW_UP -> NEGOTIATION -> CONVERTED / LOST. These exact status names are an implementation proposal; the company reference requires traceable status transitions but does not prescribe the enum values.

9. Success Metrics

- Enquiry-to-lead creation rate
- Lead-to-site-visit rate
- Site-visit-to-conversion rate
- Average first-response time
- Open leads by agent and stage
- Properties generating the most enquiries
- Converted vs. lost opportunities

10. Technology Direction

Recommended implementation: Java 17/21, Spring Boot, Spring Security with JWT, Spring Data JPA, PostgreSQL, Swagger/OpenAPI, Docker and a focused React + TypeScript web frontend. A modular monolith is preferred for the one-week MVP; microservices can be discussed as a future scaling option.

11. One-Week Delivery Plan

Day	Target
1	Market research, requirements, product scope.
2	Architecture, ERD, API contracts and project setup.
3	Authentication, users/roles and property backend.
4	Search, shortlist, enquiry and lead CRM.
5	Site visits, follow-ups and frontend integration.
6	Admin/dashboard, testing, validation and Docker.
7	Documentation, screenshots/demo, final presentation and interview rehearsal.

12. Deliverables

- Working source code repository (backend + frontend).
- README with setup/run instructions.
- Project report containing research, requirements and design decisions.
- Architecture and database ER diagrams (to be finalized on Day 2).
- Swagger/OpenAPI documentation.
- Testing evidence and screenshots.
- Final interview/demo presentation.

13. Sources

- Company-provided Real Estate Software Project reference presentation (primary assignment source).
- Magicbricks official search/help and About pages (marketplace/search capability research).
- NoBroker official site and About page (marketplace, owner listing and trust proposition research).
- Sell.Do official Real Estate CRM, Lead Management and Sales & Pre-Sales pages (CRM workflow research).